UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

# Problems on Maxima and Minima

Complete study pack for class 010: locating stationary points, classifying them with $rt - s^2$, functions with many stationary points, and applied problems solved by eliminating one variable

Compiled from the PESU class 010 slide deck, its two AV-summary transcripts, the Unit 1 class-work sheet and past ESA papers. Generated 15 October 2026.

# · How to use this pack

Topic 1 is the method and the ESA functions (every problem follows Figure F1.1 and the ladder F1.2). Topic 2 works the slides' functions with many stationary points; topic 3 the applied problems, where a constraint is used to eliminate one variable first. The theory (definitions of extrema and saddle points, the necessary conditions $f_x = f_y = 0$, the $rt - s^2$ test) is taught in class 009, which has its own pack; the test is restated here so that every problem can be solved from this pack alone. Constrained problems by Lagrange's multipliers are class 011. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S010·8 | Deck 010, slide 8 (rendered .pptx) |
| T010a 00:07:38 | AV-summary transcript “Extremas of functions continued” (older lecture), segment start |
| T010b 00:03:12 | AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture) |
| CW U1 P29 | Unit 1 class-work problem 29 |
| ESA Dec-23 1c | Past paper: Dec 2023, question 1.c |
| ESA Jan-24 s2 1a | Past paper: Jan 2024 set 2, question 1.a |

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1. 1 · The method, and the ESA cubic functions
2. 2 · Functions with many stationary points
3. 3 · Applied problems: eliminate one variable first
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

# 1 The method, and the ESA cubic functions

Every problem follows one routine: find $f_x$ and $f_y$, solve $f_x = f_y = 0$ for the stationary points (factorise one equation and split into cases), compute $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$, and classify each point by $rt - s^2$ and $r$. Both ESA functions are cubics with four stationary points: one maximum, one minimum and two saddle points. Always finish by substituting each extremum into $f$ for the value.

### Key terms and definitions

**Stationary (critical) points.** The solutions of $f_x = 0$, $f_y = 0$, solved simultaneously; the points at which the function may have an extreme value. S010·7 · T010a 00:04:13 · T010b 00:01:23

**Maximum / minimum value.** The value $f(a, b)$ at a point of maximum or minimum, obtained by substituting the point into $f$. S010·8 · T010b 00:04:02

> *"If $rt - s^2 > 0$ and $r < 0$ at $(a, b)$, $f(a, b)$ is a maximum value; if $rt - s^2 > 0$ and $r > 0$, a minimum value; if $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. $(a, b)$ is a saddle point; if $rt - s^2 = 0$ the case is doubtful and needs further investigation (class 009, restated)."* S010·8 · T010b 00:05:00

### Formulas

**E1.1 · Stationary points** ESA ★★★

$$f_x = 0, \qquad f_y = 0 \quad \text{(solve simultaneously)}$$

| | |
|---|---|
| gives | All candidate points. |
| applies | Every problem, first step. |
| note | Factorise: $6xy - 30y = 6y(x - 5) = 0$ gives the cases $y = 0$ or $x = 5$. |

S010·7 · T010b 00:01:23 · T010b 00:01:40

**E1.2 · The test** ESA ★★★

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}$$
$$rt - s^2 > 0 :\ r < 0 \text{ max},\ r > 0 \text{ min}; \qquad rt - s^2 < 0 :\ \text{saddle}; \qquad rt - s^2 = 0 :\ \text{doubtful}$$

| | |
|---|---|
| gives | The nature of each stationary point. |
| applies | At each point separately. |
| note | Restated from class 009. |

S010·8 · T010b 00:03:12 · T010b 00:04:02

### E1.3 · Hessian of the ESA cubic ESA ★★★

$$f = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x: \qquad rt - s^2 = 36\left[(x-5)^2 - y^2\right]$$

GIVES $rt - s^2$ at every point at once.

APPLIES Dec-23 1c, Jan-24 s1 1a.

NOTE $r = t = 6x - 30$, $s = 6y$. Slide 010·8 drops the 36 on $y^2$.

S010·8 · T010b 00:03:30

## Graphs and diagrams

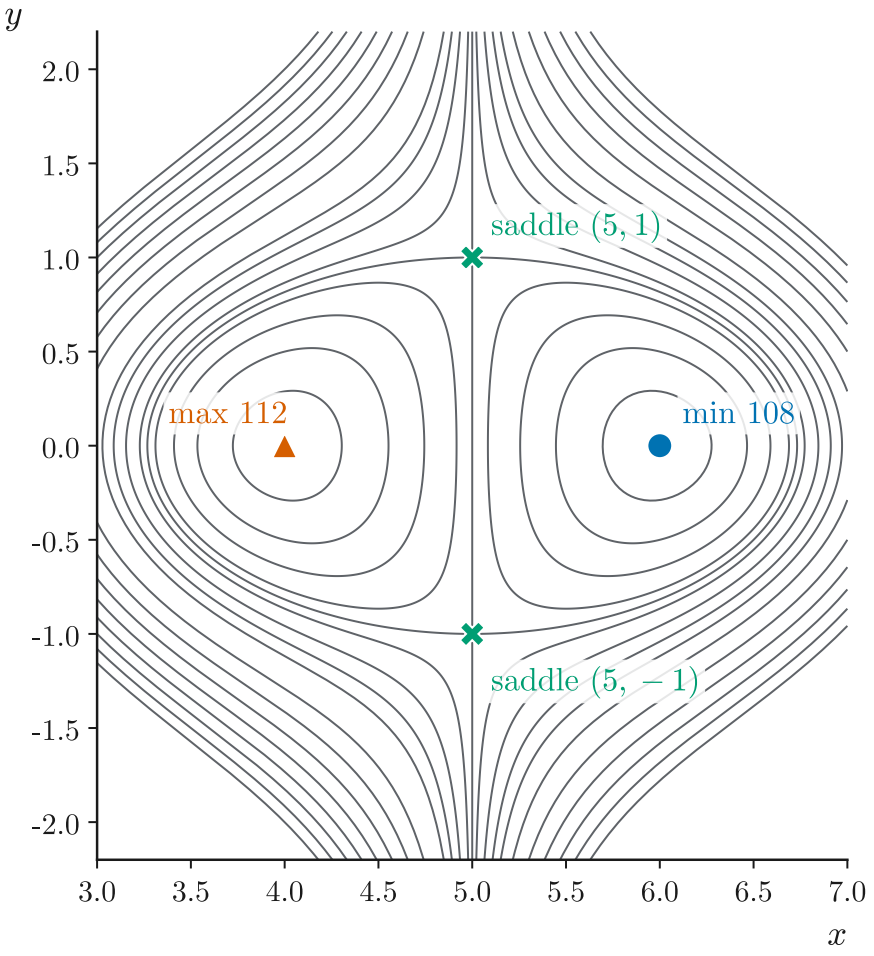



**Figure G1.1.** Level curves of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. ESA ★★★

ADDED FOR CLARITY S010·8 · ESA Dec-23 1c · ESA Jan-24 s1 1a

***How to read it.*** Grey lines are level curves $f =$ constant. Around the maximum $(4, 0)$ and the minimum $(6, 0)$ they close into small loops; at the saddle points $(5, \pm 1)$ two level curves cross, the sign of the picture of a saddle.

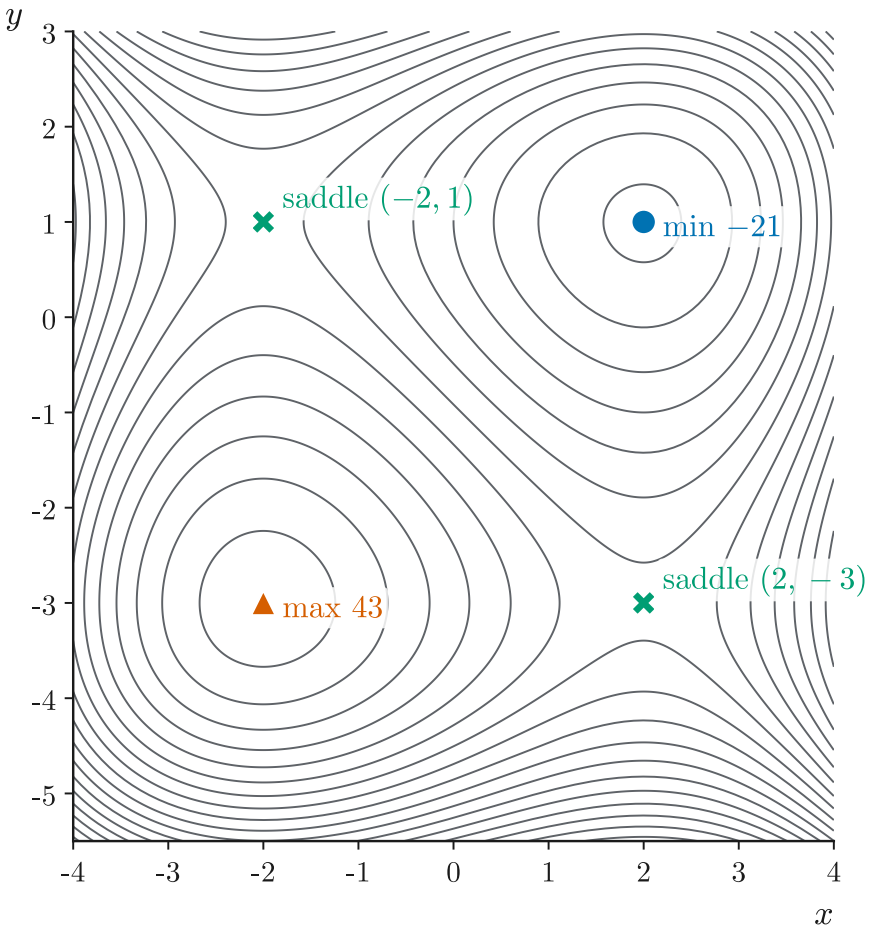
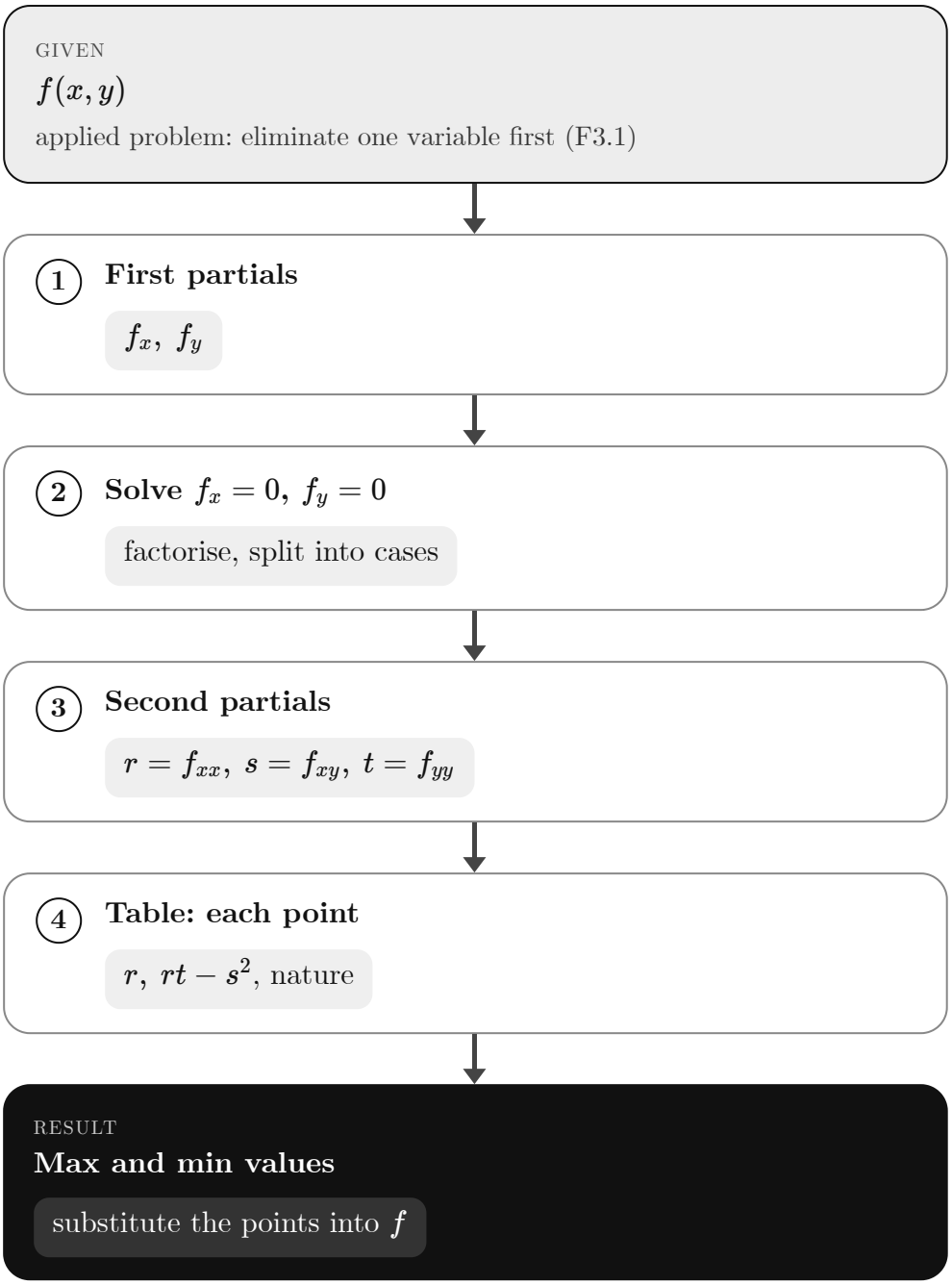



**Figure G1.2.** Level curves of $x^3 - 12x + y^3 + 3y^2 - 9y$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 1a · CW U1 P29

***How to read it.*** The four stationary points sit at the corners of a rectangle, $x = \pm 2$, $y = 1, -3$. Closed loops surround the maximum $(-2, -3)$ and the minimum $(2, 1)$; curves cross at the saddles $(2, -3)$ and $(-2, 1)$.

## Flowcharts




***Remark.*** Write a table with one row per stationary point: point, $r$, $s$, $t$, $rt - s^2$, nature, value.

**Figure F1.1.** Solving a maxima–minima problem ESA ★★★

S010·7 · S010·8 · T010b 00:05:00

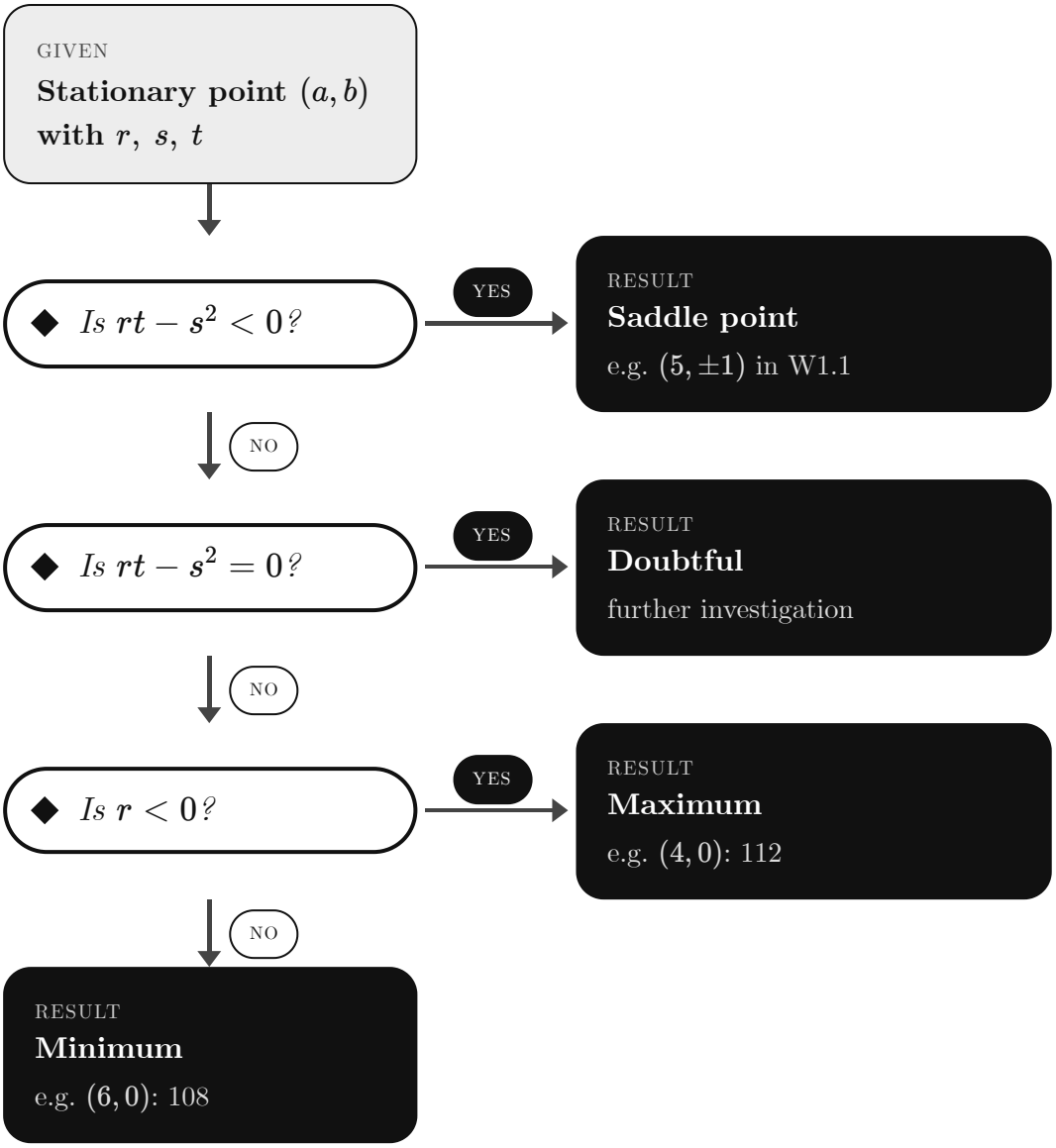



**Figure F1.2.** Classifying each stationary point ESA ★★★

S010·8 · T010b 00:04:02

### Stationary points of the two ESA functions

| Function | Point | $r$ | $rt - s^2$ | Nature | $f$ |
|---|---|---|---|---|---|
| $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ | $(4, 0)$ | $-6$ | $36$ | Maximum | $112$ |
| | $(6, 0)$ | $6$ | $36$ | Minimum | $108$ |
| | $(5, \pm 1)$ | $0$ | $-36$ | Saddle | — |
| $x^3 - 12x + y^3 + 3y^2 - 9y$ | $(-2, -3)$ | $-12$ | $144$ | Maximum | $43$ |
| | $(2, 1)$ | $12$ | $144$ | Minimum | $-21$ |
| | $(2, -3)$, $(-2, 1)$ | $\pm 12$ | $-144$ | Saddle | — |

S010·8 · ESA Dec-23 1c · ESA Jan-24 s2 1a

## Important points and why

1. Procedure (lecture): find $p = f_x$, $q = f_y$; equate both to zero; find the critical points; compute $r$, $s$, $t$ ; evaluate $rt - s^2$ at each critical point; conclude. ESA ★★★ T010b 00:05:00 · T010b 00:05:16
2. Split $f_y = 0$ into cases and substitute each into $f_x = 0$: for the ESA cubic, $y = 0$ gives $x = 4, 6$ and $x = 5$ gives $y = \pm 1$. ESA ★★★ S010·7 · S010·8 · T010b 00:01:40
3. Count the stationary points before classifying; a cubic in $x$ and $y$ typically gives four, a quartic up to nine. ESA ★★ ADDED FOR CLARITY S010·8 · S010·11
4. A saddle point has no value to report; give the maximum and minimum values only. ESA ★★ S010·8
5. When $rt - s^2 > 0$, $r$ and $t$ have the same sign, so either decides between maximum and minimum. ESA ★ ADDED FOR CLARITY S010·8

**Why does $f_y = 6y(x - 5) = 0$ give two separate cases?**
A product is zero when either factor is zero: $y = 0$ or $x = 5$. Each case is substituted into $f_x = 0$ to find its own points; missing a case loses stationary points.

ADDED FOR CLARITY S010·7 · T010b 00:01:40

***Exam trap.*** Slide 010·8 says the maximum value is at $(6, 0)$; the maximum 112 is at $(4, 0)$, the minimum 108 at $(6, 0)$. S010·8 · T010b 00:04:42

***Exam trap.*** $rt - s^2 = 36(x - 5)^2 - 36y^2$: the 36 multiplies $y^2$ too (slide 010·8 drops it).

ADDED FOR CLARITY S010·8

## Worked examples

### W1.1 · Extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ ESA ★★★

**given** Find the extreme values (locate the stationary points and discuss their nature) of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

**step 1** $f_x = 3x^2 + 3y^2 - 30x + 72$, $f_y = 6xy - 30y = 6y(x - 5)$.

**step 2** $f_y = 0$: $y = 0$ or $x = 5$. If $y = 0$: $3x^2 - 30x + 72 = 0$, $x^2 - 10x + 24 = 0$, $x = 4$ or $6$. If $x = 5$: $75 + 3y^2 - 150 + 72 = 0$, $y^2 = 1$, $y = \pm 1$.

**step 3** Stationary points: $(4, 0)$, $(6, 0)$, $(5, 1)$, $(5, -1)$.

**step 4** $r = 6x - 30$, $s = 6y$, $t = 6x - 30$, so $rt - s^2 = 36[(x - 5)^2 - y^2]$.

**step 5** $(4, 0)$: $r = -6$, $rt - s^2 = 36 > 0$: maximum, $f = 64 - 240 + 288 = 112$.

**step 6** $(6, 0)$: $r = 6$, $rt - s^2 = 36 > 0$: minimum, $f = 216 - 540 + 432 = 108$.

**step 7** $(5, \pm 1)$: $r = 0$, $s = \pm 6$, $rt - s^2 = -36 < 0$: saddle points.

**answer** Maximum $112$ at $(4, 0)$; minimum $108$ at $(6, 0)$; $(5, 1)$ and $(5, -1)$ are saddle points

**check** CW P32 answer agrees; sympy finds the same four points and values. Slide 010·8's "maximum ... at $(6, 0)$" is a slip for $(4, 0)$.

S010·7 · S010·8 · T010b 00:00:23 · ESA Dec-23 1c · ESA Jan-24 s1 1a · CW U1 P32

### W1.2 · Stationary points of $x^3 - 12x + y^3 + 3y^2 - 9y$ ESA ★★★

**given** Locate the stationary points of $f(x, y) = x^3 - 12x + y^3 + 3y^2 - 9y$ and discuss their nature.

**step 1** $f_x = 3x^2 - 12 = 0 \Rightarrow x = \pm 2$; $f_y = 3y^2 + 6y - 9 = 3(y + 3)(y - 1) = 0 \Rightarrow y = 1, -3$.

**step 2** Stationary points: $(2, 1)$, $(2, -3)$, $(-2, 1)$, $(-2, -3)$.

**step 3** $r = 6x$, $s = 0$, $t = 6y + 6$; $rt - s^2 = 36x(y + 1)$.

**step 4** $(2, 1)$: $r = 12$, $rt - s^2 = 144 > 0$: minimum, $f = 8 - 24 + 1 + 3 - 9 = -21$.

**step 5** $(-2, -3)$: $r = -12$, $rt - s^2 = 144 > 0$: maximum, $f = -8 + 24 - 27 + 27 + 27 = 43$.

**step 6** $(2, -3)$ and $(-2, 1)$: $rt - s^2 = -144 < 0$: saddle points.

**answer** Maximum $43$ at $(-2, -3)$; minimum $-21$ at $(2, 1)$; $(2, -3)$ and $(-2, 1)$ are saddle points

**check** CW P29 answer agrees; sympy agrees.

ESA Jan-24 s2 1a · CW U1 P29

## Past-paper questions

**ESA Dec 2023** 7 marks UE22MA141A Q1.c · solved in W1.1

Find the extreme values of the function $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

**ESA Jan 2024** 7 marks UE23MA141A set 1, Q1.a · solved in W1.1

Locate the stationary points of $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. Discuss their nature.

**ESA Jan 2024** 7 marks UE23MA141A set 2, Q1.a · solved in W1.2

Locate the stationary points of $f(x,y) = x^3 - 12x + y^3 + 3y^2 - 9y$. Discuss their nature.

SHORT-NOTE RECAP

$f_x = 0$, $f_y = 0$ → split into cases → stationary points → $r, s, t$ → $rt - s^2$ and $r$ → values. ESA: $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4,0)$, min 108 at $(6,0)$, saddles $(5, \pm 1)$. $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2,-3)$, min $-21$ at $(2,1)$, saddles $(2,-3)$, $(-2,1)$.

# 2 Functions with many stationary points

The slides test two more functions. $x^4 + y^4 - x^2 - y^2 + 1$ has nine stationary points (a maximum at the origin, four minima and four saddle points). $\sin x + \sin y + \sin(x + y)$ needs trigonometric equations: subtracting the two conditions gives $x = y$, then $\cos 2x = -\cos x$ gives $x = \frac{\pi}{3}$, a maximum $\frac{3\sqrt{3}}{2}$.

### Graphs and diagrams

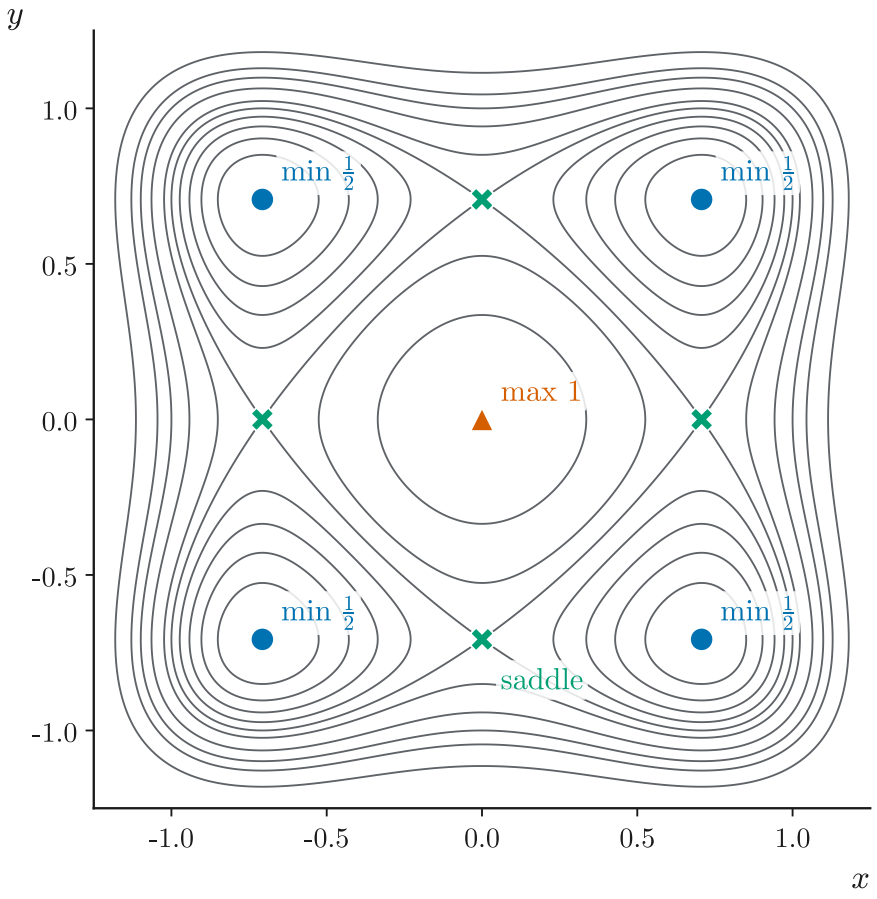



**Figure G2.1.** Level curves of $x^4 + y^4 - x^2 - y^2 + 1$. ESA ★★

ADDED FOR CLARITY S010·11 · S010·12

***How to read it.*** Nine stationary points on a $3 \times 3$ grid: a maximum at the centre, minima at the four corners $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$ and saddles at the four edge midpoints on the axes.

### Important points and why

1. Separable functions such as $x^4 + y^4 - x^2 - y^2 + 1$ give $f_x$ in $x$ only and $f_y$ in $y$ only: combine every root of one with every root of the other. ESA ★★ S010·11 · T010b 00:08:29
2. For $\sin x + \sin y + \sin(x + y)$, subtract the two conditions to get $\cos x = \cos y$, i.e. $x = y$ in the working range; then $\cos 2x = -\cos x = \cos(\pi - x)$ gives $x = \frac{\pi}{3}$. ESA ★★ S010·13 · T010b 00:10:23

***Exam trap.*** When $rt - s^2 < 0$ the sign of $r$ is irrelevant; at $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ (not $-2$ as slide 010·12 says), and the point is still a saddle. ADDED FOR CLARITY S010·12 · T010b 00:09:26

***Exam trap.*** At $(\frac{\pi}{3}, \frac{\pi}{3})$, $s = -\frac{\sqrt{3}}{2}$, so $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$).
ADDED FOR CLARITY S010·14

## Worked examples

### W2.1 · $x^4 + y^4 - x^2 - y^2 + 1$: nine stationary points — ESA ★★

**GIVEN** Test $f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$ for maxima, minima and saddle points.

**STEP 1** $f_x = 4x^3 - 2x = 2x(2x^2 - 1) = 0 \Rightarrow x = 0, \pm\frac{1}{\sqrt{2}}$; likewise $y = 0, \pm\frac{1}{\sqrt{2}}$: nine points.

**STEP 2** $r = 12x^2 - 2$, $s = 0$, $t = 12y^2 - 2$.

**STEP 3** $(0, 0)$: $r = t = -2$, $rt - s^2 = 4 > 0$, $r < 0$: maximum $f = 1$.

**STEP 4** $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$: $r = t = 4$, $rt - s^2 = 16 > 0$: minima, $f = \frac{1}{4} + \frac{1}{4} - \frac{1}{2} - \frac{1}{2} + 1 = \frac{1}{2}$.

**STEP 5** $(\pm\frac{1}{\sqrt{2}}, 0)$: $r = 4$, $t = -2$; $(0, \pm\frac{1}{\sqrt{2}})$: $r = -2$, $t = 4$; both $rt - s^2 = -8 < 0$: saddle points.

**ANSWER** Maximum $1$ at $(0, 0)$; minimum $\frac{1}{2}$ at the four points $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$; four saddle points on the axes

**CHECK** sympy agrees (saddle values $\frac{3}{4}$).

S010·11 · S010·12 · T010b 00:08:12

### W2.2 · $\sin x + \sin y + \sin(x + y)$ — ESA ★★

**GIVEN** Test $f(x, y) = \sin x + \sin y + \sin(x + y)$ for maxima, minima and saddle points ( $0 < x, y < \pi$).

**STEP 1** $f_x = \cos x + \cos(x + y) = 0$ (1), $f_y = \cos y + \cos(x + y) = 0$ (2).

**STEP 2** (1) − (2): $\cos x = \cos y$, so $x = y$. Then (1): $\cos 2x = -\cos x = \cos(\pi - x)$, $2x = \pi - x$, $x = \frac{\pi}{3}$.

**STEP 3** $r = -\sin x - \sin(x + y)$, $s = -\sin(x + y)$, $t = -\sin y - \sin(x + y)$.

**STEP 4** At $(\frac{\pi}{3}, \frac{\pi}{3})$: $r = t = -\sqrt{3}$, $s = -\frac{\sqrt{3}}{2}$, $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4} > 0$, $r < 0$: maximum.

**STEP 5** $f = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$.

**ANSWER** Maximum $\dfrac{3\sqrt{3}}{2}$ at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$

**CHECK** sympy: $rt - s^2 = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$); the conclusion is the same.

S010·13 · S010·14 · T010b 00:09:50

SHORT-NOTE RECAP

$x^4+y^4-x^2-y^2+1$: max 1 at $(0,0)$, min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}},\pm\frac{1}{\sqrt{2}})$, four saddles on the axes. $\sin x+\sin y+\sin(x+y)$: $x=y=\frac{\pi}{3}$, max $\frac{3\sqrt{3}}{2}$.

# 3 Applied problems: eliminate one variable first

When the quantity depends on three variables tied by a condition (a sphere, a fixed volume), use the condition to eliminate one variable, then apply the two-variable method. The temperature $400xyz^2$ on the unit sphere is highest, 50, at $(\pm\frac{1}{2}, \pm\frac{1}{2})$ with equal signs. An open box of volume 10 m³ uses least metal when the base is $\sqrt[3]{20}$ square and the height half of that. Class 011 does such problems without eliminating.

### Key terms and definitions

**Elimination.** Using the given condition (constraint) to express one variable in terms of the others, so that the quantity becomes a function of two independent variables. T010a 00:00:40 · T010a 00:08:27

### Formulas

**E3.1 · Eliminating with a sphere** ESA ★★

$$x^2+y^2+z^2=a^2 \;\Rightarrow\; z^2=a^2-x^2-y^2, \qquad T=kxyz^2=kxy(a^2-x^2-y^2)$$

| | |
|---|---|
| gives | A function of $x, y$ only. |
| applies | Quantity contains $z^2$ and the point lies on a sphere. |
| note | Unit sphere: $a=1$. |

S010·3 · S010·9 · T010a 00:01:00

**E3.2 · Open box of volume V** ESA ★★

$$A=xy+2yz+2zx, \quad xyz=V \;\Rightarrow\; A=xy+\frac{2V}{x}+\frac{2V}{y}$$

| | |
|---|---|
| gives | Surface area (metal used) of a box with no top, base $x \times y$. |
| applies | Least material for a given volume. |
| note | Minimum at $x=y=\sqrt[3]{2V}$, $z=\frac{x}{2}$ (CW P34). The slide's $A=2xy+2yz+zx$ is not the area of an open box. |

added for clarity S010·5 · CW U1 P34

## Graphs and diagrams

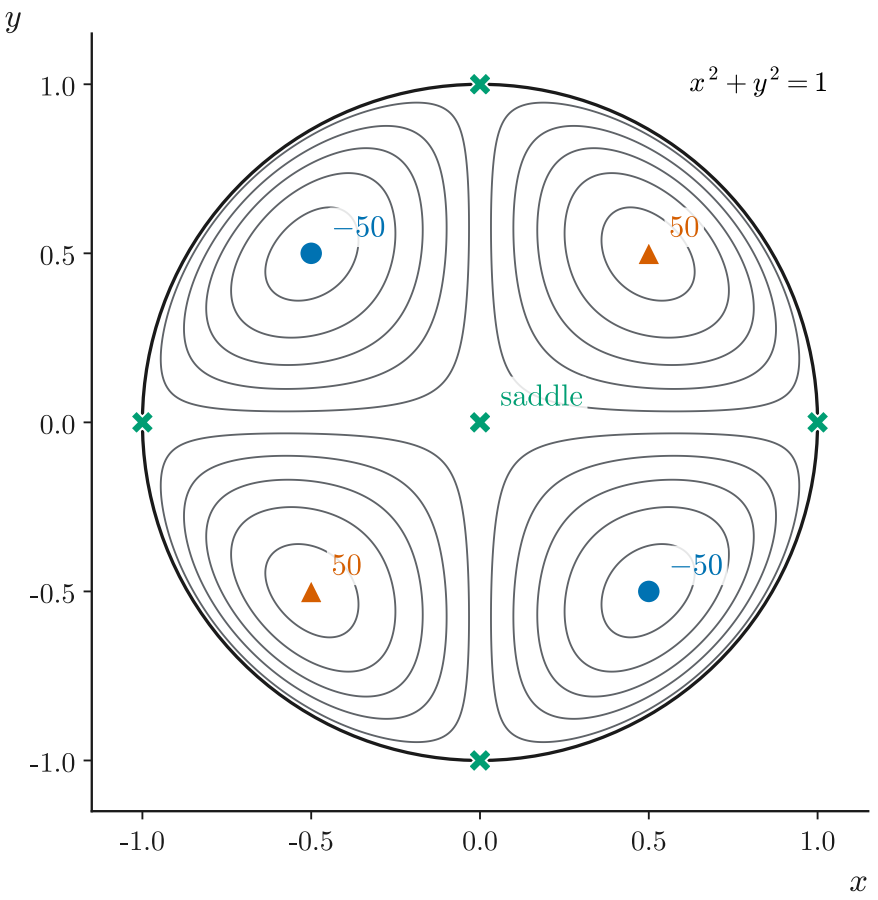



**Figure G3.1.** Temperature $400xy(1 - x^2 - y^2)$ over the unit disk. ESA ★★

ADDED FOR CLARITY S010·3 · S010·4

***How to read it.*** Inside the unit circle (the projection of the sphere) the temperature is positive in the first and third quadrants and negative in the others. It peaks at $50$ at $(\tfrac{1}{2}, \tfrac{1}{2})$ and $(-\tfrac{1}{2}, -\tfrac{1}{2})$ and is lowest, $-50$, at the other two. The origin and $(\pm 1, 0)$, $(0, \pm 1)$, where $T = 0$, are saddles.

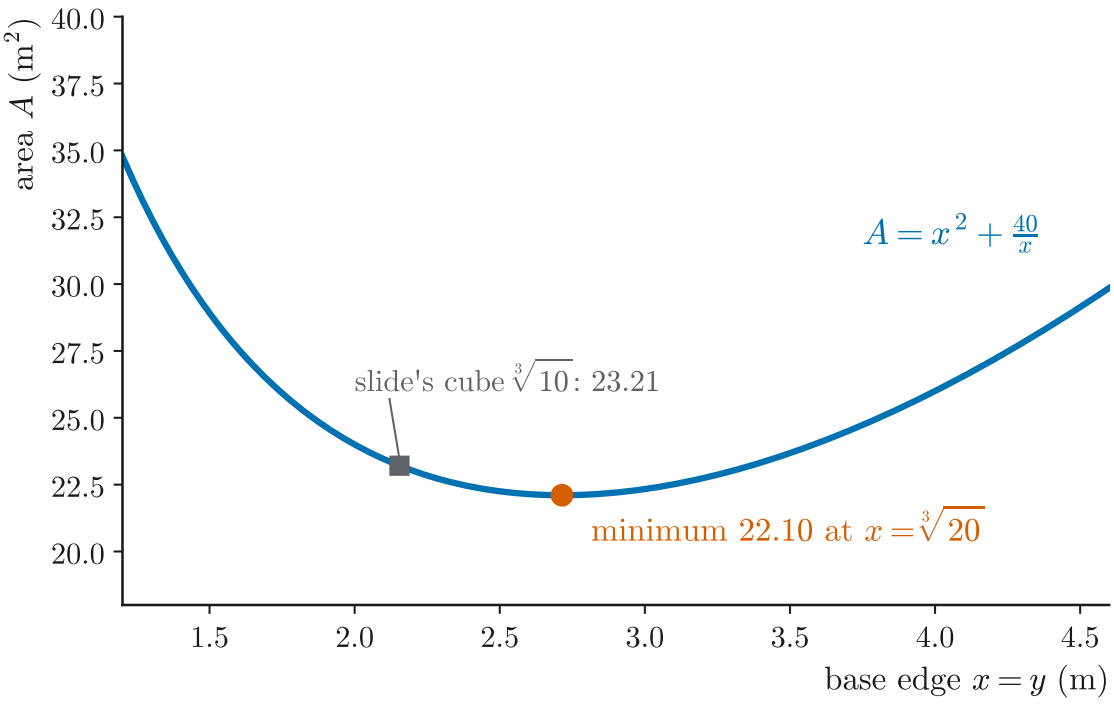
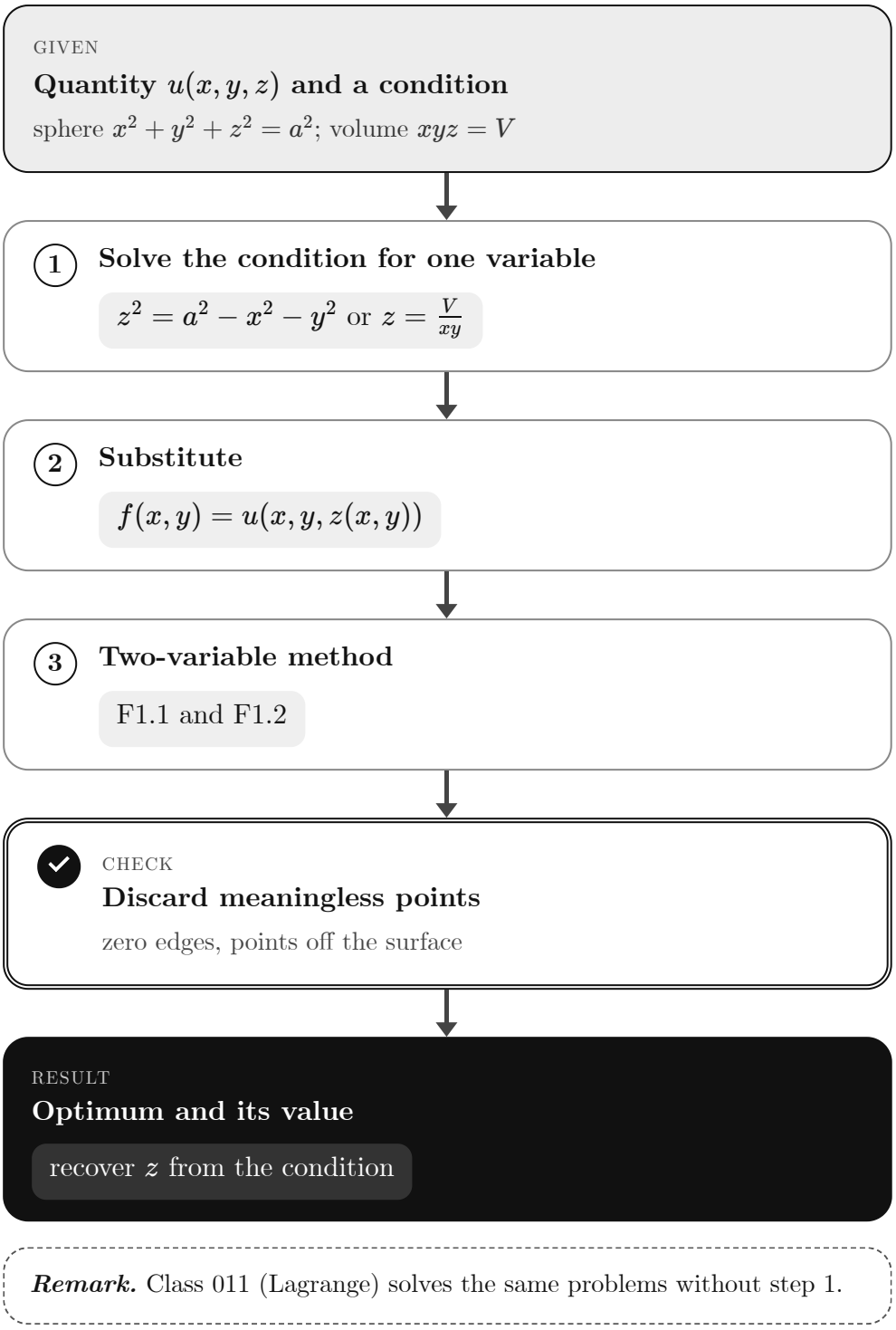



**Figure G3.2.** Metal used by an open box of 10 m³ with a square base. ESA ★

ADDED FOR CLARITY S010·6 · CW U1 P34

***How to read it.*** Along $x = y$ the area is $A = x^2 + \frac{40}{x}$ (blue). Its minimum, **22.10** m², is at $x = \sqrt[3]{20} \approx 2.71$ m. The slide's cube, $x = \sqrt[3]{10} \approx 2.15$ m, sits higher on the curve (**23.21** m²).

## Flowcharts




***Remark.*** Class 011 (Lagrange) solves the same problems without step 1.

**Figure F3.1.** Applied problem by elimination ESA ★★

S010·3 · S010·5 · T010a 00:00:40 · T010a 00:08:27

## Important points and why

1. We cannot yet find extrema of a function of three variables, so use the constraint to eliminate one (here $z^2$ or $z$). ESA ★★ T010a 00:00:40 · T010a 00:08:27

2. A product such as $kxyz^2$ is largest when $x$ and $y$ have the same sign and $k > 0$; points with $x$, $y$ of opposite signs give the lowest temperature. ESA ★★ S010·10 · T010b 00:05:40
3. Reject stationary points that make no physical sense (an edge $y = 0$ in the box problem makes $x \to \infty$). ESA ★ T010a 00:13:07 · T010a 00:13:48
4. Elimination is not always easy; class 011's Lagrange multipliers avoid it (T010a 00:16:49). ESA ★ T010a 00:16:49

**Why is a cube the wrong answer for an open box?**
Without a top the base is used once and each side face twice in the area, so the box should be lower and wider: $x = y = \sqrt[3]{20}$, $z = \frac{1}{2}\sqrt[3]{20}$. The cube $x = y = z = \sqrt[3]{V}$ minimises the area of a closed box $2(xy + yz + zx)$.
ADDED FOR CLARITY S010·6 · CW U1 P34

***Exam trap.*** $400xy(1 - x^2 - y^2)$ also has stationary points $(\pm 1, 0)$, $(0, \pm 1)$ (saddles) that the slides omit; and slide 010·4's $rt - s^2$ values are $-160000$ and $320000$, not $-1600$ and $356000$.
ADDED FOR CLARITY S010·3 · S010·4

***Exam trap.*** Open box: area $xy + 2yz + 2zx$ (one base, four sides). The slide's $2xy + 2yz + zx$ and its derivative slip lead to a cube, which is wrong. ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:38

## Worked examples

### W3.1 · Highest temperature $400xyz^2$ on the unit sphere — ESA ★★★

**GIVEN** The temperature at $(x, y, z)$ is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

**STEP 1** Eliminate $z^2 = 1 - x^2 - y^2$: $f(x, y) = 400xy(1 - x^2 - y^2) = 400(xy - x^3y - xy^3)$.

**STEP 2** $f_x = 400y(1 - 3x^2 - y^2) = 0$, $f_y = 400x(1 - x^2 - 3y^2) = 0$.

**STEP 3** Both brackets zero: subtracting gives $x^2 = y^2$, then $4x^2 = 1$: $(\pm\frac{1}{2}, \pm\frac{1}{2})$. Also $(0, 0)$, $(\pm 1, 0)$, $(0, \pm 1)$.

**STEP 4** $r = -2400xy$, $s = 400(1 - 3x^2 - 3y^2)$, $t = -2400xy$.

**STEP 5** $(\frac{1}{2}, \frac{1}{2})$, $(-\frac{1}{2}, -\frac{1}{2})$: $r = -600$, $s = -200$, $rt - s^2 = 320000 > 0$, $r < 0$: maxima, $T = 400 \cdot \frac{1}{4} \cdot \frac{1}{2} = 50$.

**STEP 6** $(0, 0)$: $rt - s^2 = -160000 < 0$; $(\pm 1, 0)$, $(0, \pm 1)$: $-640000 < 0$: saddles. Opposite signs: minima, $T = -50$.

**ANSWER** Highest temperature $50$, at $x = y = \pm\frac{1}{2}$, $z = \pm\frac{1}{\sqrt{2}}$

**CHECK** sympy agrees. ESA June 2024 Q1.d asks this by Lagrange's method (class 011 pack) and gets the same 50.

S010·3 · S010·4 · T010a 00:00:00 · T010a 00:06:33 · ESA Jun-24 1d

### W3.2 · Highest temperature $kxyz^2$ on a sphere of radius $a$ ESA ★★

**GIVEN** $T = kxyz^2$ ($k > 0$). Find the highest temperature on $x^2 + y^2 + z^2 = a^2$.

**STEP 1** $T = kxy(a^2 - x^2 - y^2)$; $T_x = ky(a^2 - 3x^2 - y^2)$, $T_y = kx(a^2 - x^2 - 3y^2)$.

**STEP 2** Stationary points: $(0,0)$ and $(\pm\frac{a}{2}, \pm\frac{a}{2})$ (also $(\pm a, 0)$, $(0, \pm a)$, saddles).

**STEP 3** $r = -6kxy$, $s = k(a^2 - 3x^2 - 3y^2)$, $t = -6kxy$.

**STEP 4** $(0,0)$: $r = 0$, $rt - s^2 = -k^2a^4 < 0$: saddle.

**STEP 5** $(\frac{a}{2}, \frac{a}{2})$, $(-\frac{a}{2}, -\frac{a}{2})$: $r = -\frac{6ka^2}{4} < 0$, $s = -\frac{ka^2}{2}$, $rt - s^2 = \frac{9k^2a^4}{4} - \frac{k^2a^4}{4} = 2k^2a^4 > 0$: maximum.

**ANSWER** $$\boxed{T_{\max} = k \cdot \frac{a}{2} \cdot \frac{a}{2} \cdot \frac{a^2}{2} = \frac{ka^4}{8}}$$

**CHECK** With $k = 400$, $a = 1$: $50$, as in W3.1; sympy agrees.

S010·9 · S010·10 · T010b 00:05:25

### W3.3 · Open box of 10 m³ with least metal ESA ★★

**GIVEN** A container with an open top is to have 10 m³ capacity and be made of thin sheet metal. Find the dimensions that use the least metal.

**STEP 1** Base $x \times y$, height $z$: $A = xy + 2yz + 2zx$, $xyz = 10$, so $A = xy + \frac{20}{x} + \frac{20}{y}$.

**STEP 2** $A_x = y - \frac{20}{x^2} = 0$, $A_y = x - \frac{20}{y^2} = 0$; then $x^2y = xy^2 = 20$, so $x = y$ and $x^3 = 20$: $x = y = \sqrt[3]{20}$.

**STEP 3** $z = \frac{10}{xy} = \frac{10}{20^{2/3}} = \frac{1}{2}\sqrt[3]{20}$.

**STEP 4** $A_{xx} = \frac{40}{x^3} = 2$, $A_{xy} = 1$, $A_{yy} = \frac{40}{y^3} = 2$: $rt - s^2 = 3 > 0$, $r > 0$: minimum.

**ANSWER** $\boxed{x = y = \sqrt[3]{20} \approx 2.71 \text{ m}, \text{ height } z = \frac{1}{2}\sqrt[3]{20} \approx 1.36 \text{ m}; \text{ least area } 3 \cdot 20^{2/3} \approx 22.10 \text{ m}^2}$

**CHECK** Matches CW P34 ($x = y = (2V)^{1/3}$, $z = x/2$); sympy agrees. The slide's cube of side $\sqrt[3]{10}$ needs $23.21$ m², more metal.

ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:06 · CW U1 P34

## Past-paper questions

ESA June 2024 · 7 marks · UE23MA141A Q1.d · solved in W3.1

The temperature $T$ at any point $(x, y, z)$ in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (Asked by Lagrange's method; W3.1 checks the answer by elimination.)

SHORT-NOTE RECAP

Eliminate with the condition, then use the two-variable test. $400xyz^2$ on the unit sphere: max 50 at $(\pm\frac{1}{2},\pm\frac{1}{2})$ (same signs); $kxyz^2$ on radius $a$: $\frac{ka^4}{8}$. Open box, volume $V$: $A = xy + \frac{2V}{x} + \frac{2V}{y}$, $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$.

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|
| $p$, $q$ | First partials | $p = f_x$, $q = f_y$ (lecture T010b) | E1.1 |
| $r$, $s$, $t$ | Second partials | $f_{xx}$, $f_{xy}$, $f_{yy}$ at a point | E1.2, E1.3 |
| $rt - s^2$ | Hessian | Decides extremum or saddle | E1.2, E1.3 |
| $T$ | Temperature | $400xyz^2$ or $kxyz^2$; not the second partial $t$ | E3.1 |
| $k$ | Constant | Positive constant in $T = kxyz^2$ | E3.1 |
| $a$ | Radius | Sphere $x^2 + y^2 + z^2 = a^2$ | E3.1 |
| $A$ | Surface area | Metal used by the open box | E3.2 |
| $V$ | Volume | Fixed capacity $xyz = V$ (10 m³) | E3.2 |

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Deck 010 (.pptx rendered; .slides.md; .slides.txt) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.* | 15 slides; content on 3–14 |
| Transcript T010a (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.* | 17.3 min: slides 3–6 |
| Transcript T010b (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.* | 11.1 min: slides 7–14 |
| Unit 1 & 2 class-work sheet (UE24MA141A) | pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf | Problems 29, 31, 32 (with answers); 30 is blank; 34 confirms the open-box answer |
| ESA papers, UE22MA141A and UE23MA141A (8 papers) | pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf | 3 in-scope Q1 parts (2 distinct functions) |
| Videos 010 (.mp4) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_*.mp4 | Not opened: every slide is readable and every disputed number was settled by recomputing |

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Free extrema of $f(x, y)$ are asked in three of them, always a 7-mark Q1 part: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a ask the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ (also slide 010·7 and CW problem 32), and Jan 2024 set 2 Q1.a asks $x^3 - 12x + y^3 + 3y^2 - 9y$ (CW problem 29). Both are cubic polynomials with four stationary points: one maximum, one minimum and two saddle points. The other five papers ask Lagrange's method (class 011) instead. The temperature-on-a-sphere problem of slide 010·3 is asked in June 2024 Q1.d, but by Lagrange's method; the elimination method of topic 3 gives the same answer. Trigonometric and quartic functions (topic 2) were never asked.

### Known gaps, conflicts and corrections

1. Slide 010·4 prints $rt - s^2 = -1600$ at $(0, 0)$ and $356000$ at $(\pm\frac{1}{2}, \pm\frac{1}{2})$ for $400xy(1 - x^2 - y^2)$. The correct values are $-160000$ and $320000$ (sympy; also the special case $k = 400$, $a = 1$ of slide 010·10's $-k^2a^4$ and $2k^2a^4$). The conclusions (saddle, maximum 50) are unaffected.
2. Slides 010·3–4 and 010·9 miss four stationary points: $(\pm 1, 0)$ and $(0, \pm 1)$ (generally $(\pm a, 0)$, $(0, \pm a)$), from $y = 0$, $1 - x^2 = 0$ and $x = 0$, $1 - y^2 = 0$. They are saddle points with $T = 0$ (on the circle $z = 0$). The slides also list $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$ but never classify them: they are minima of $T$ ($-50$, the lowest temperature).

3. Slides 010·5–6 and the lecture (T010a 00:10:33) differentiate $A = 2xy + \frac{20}{x} + \frac{10}{y}$ wrongly: $A_y = 2x - \frac{10}{y^2}$, not $2x - \frac{20}{y^2}$, and $A_{yy} = \frac{20}{y^3}$, not $\frac{40}{y^3}$. The printed answer, a cube of side $\sqrt[3]{10}$, is therefore wrong (area $23.21$). The correct box has the two edges of the open face $\sqrt[3]{20} \approx 2.714$ m and the third edge half of that, $1.357$ m (area $3 \cdot 20^{2/3} \approx 22.10$ m²), which agrees with CW problem 34 ( $x = y = (2V)^{1/3}$, $z = x/2$). A cube is the answer for a closed box.
4. Slide 010·5 measures the edges in "ft" though the capacity is 10 m³; metres are used here.
5. Slide 010·8 writes $rt - s^2 = 36(x-5)^2 - y^2$; it is $36(x-5)^2 - 36y^2$ (the values $\pm 36$ printed below use the correct form). It also says "the maximum value of $f$ at $(6, 0)$ is 112"; the maximum is at $(4, 0)$ (the lecture repeats the slip, T010b 00:04:42).
6. Slide 010·12 gives $r = -2 < 0$ for all four points $(\pm\frac{1}{\sqrt{2}}, 0)$, $(0, \pm\frac{1}{\sqrt{2}})$. At $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ and $t = -2$ ; the saddle conclusion ($rt - s^2 = -8 < 0$) is unaffected. The lecture adds "but $r$ is also negative" (T010b 00:09:26); the sign of $r$ plays no part when $rt - s^2 < 0$.
7. Slide 010·14 prints $rt - s^2 = \frac{3}{4}$ at $(\frac{\pi}{3}, \frac{\pi}{3})$ for $\sin x + \sin y + \sin(x+y)$; it is $3 - \frac{3}{4} = \frac{9}{4}$ ($s = -\frac{\sqrt{3}}{2}$). The conclusion (maximum $\frac{3\sqrt{3}}{2}$) is unaffected. The slide states no domain; "the only critical point" holds for $0 < x, y < \pi$ (added for clarity).
8. Slide 010·9 prints the sphere as $x^2 + y^2 + z^2 = a'$; it is $a^2$ (slide 010·9's next line and T010b 00:05:40).
9. The lecture T010a says "$x$ will be plus or minus 1" (00:02:42) and then corrects itself to $x = \pm y$; and "this should be $x$ equal to 0, not $y$" (00:03:27) for slide 010·3's $f_y$ line, which indeed should read $x = 0$.
10. Class 009 (definitions, necessary conditions, the $rt - s^2$ test) is a separate pack; the test is restated here without proof. Class 011 (Lagrange's multipliers) is also separate; the elimination method of topic 3 is what class 010 uses.
11. Transcript recognition errors were corrected silently: "seventy p" → 72, "minus pi square" → $-36y^2$, "115" → 150, "phi" / "5 comma" spoken as five, "plus R minus" → plus or minus, "stattle points" / "settle point" → saddle point(s), "6 comero is 12" → (4, 0) is 112, "RT when C square" → $rt - s^2$, "5 by 3" → $\frac{\pi}{3}$, "P of x y z" → $T(x, y, z)$, "minus x square y square" → $-x^2 - y^2$, "Subtracting 2 from 2" → (2) from (1), "equity" → equation, "dou" → $\partial$.

## Coverage

| Source | Citations | Slides cited (slide × citations) |
|---|---|---|
| S010 | 52 | 3×5, 4×3, 5×4, 6×4, 7×6, 8×15, 9×2, 10×2, 11×4, 12×3, 13×2, 14×2 |
| CW | 7 | |
| ESA | 9 | |
| T010a | 15 | |
| T010b | 23 | |

Slides 010·1, 010·2 and 010·15 are title, session and closing slides. Lecture T010a (older, 17.3 min) works slides 3–6 (temperature on the unit sphere, open box); lecture T010b (newer, 11.1 min) works slides 7–14. Slide 010·2 says "Session

: 9”; the bucket numbers the class 010.